HACKATHON BRIEFING | TECHNICAL WHITEPAPER | PRODUCTION READY | NEXT.JS 16 + REACT 19 + PYTHON MCTS

Smart Supply Chain Analyst (SSCA)

Global Multimodal Supply Chain Intelligence & Monte Carlo Disruption Simulation Digital Twin

PROJECT TRACK
AI, Logistics & Digital Twins

CORE AI ARCHITECTURE
Groq LLaMA 3.1 + Gemini RAG

SIMULATION ENGINE
Python FastAPI MCTS Oracle

TARGET JURISDICTIONS
Global (Singapore, India, Hubs)

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

Global supply chains are non-linear, multi-tier dependency networks governed by physical bottlenecks. Over **80% of global merchandise trade** by volume moves across maritime sea lanes, funneling through ultra-vulnerable geopolitical chokepoints including the **Strait of Malacca (90,000+ vessels/year)**, **Strait of Hormuz (20.5M barrels of oil/day)**, **Suez Canal**, and **Bab-el-Mandeb**.

> **The Core Problem Today:** When a geopolitical conflict flares, an export ban is announced, or a chokepoint is blockaded, trade analysts and logistics directors react *days or weeks too late*. Existing solutions are either **dumb news aggregators** that create alert fatigue without modeling supply-chain repercussions, or **static spreadsheets** that fail to simulate cascading multi-hop ripple effects.

A shock in the Strait of Hormuz does not merely increase tanker transit times; it cascades deterministically into **crude import deficits**, **refinery throughput throttling**, **petrochemical feedstock shortages**, **pharmaceutical API halts**, and **power grid brownouts**.

Our Solution: Smart Supply Chain Analyst (SSCA)

SSCA is an **enterprise-grade supply chain digital twin and predictive intelligence platform**. It fuses live maritime vessel tracking (AIS), real-time geopolitical OSINT feeds, live commodity spot prices, a 100+ node strategic trade knowledge graph, and an asynchronous **Python-powered Monte Carlo Tree Search (MCTS)** simulation microservice into a unified, high-performance operational cockpit.

> **Key Innovation:** Fusing deterministic graph theory (BFS topological propagation) with stochastic probabilistic modeling (MCTS Markov state rollouts) and agentic multi-LLM RAG synthesis (Groq + Gemini).

2. SYSTEM ARCHITECTURE & TECH STACK

SSCA utilizes a high-throughput **dual-stack microservice architecture** designed for sub-second reactive intelligence and zero downtime:

LAYER	TECHNOLOGY & FRAMEWORK	CORE RESPONSIBILITY
Frontend & UI Twin	Next.js 16.2 (App Router), React 19, TypeScript 5, Tailwind CSS 4, Framer Motion 12	Enterprise UI, interactive digital twin cockpit, glassmorphic telemetry panels, live charts.
Geospatial Map	MapLibre GL 5.24, TopoJSON Client, WebGL	Live global maritime map rendering 100+ vessels, trade corridors, risk heatmaps, port anchorages.
Simulation Engine	Python 3.10+, FastAPI, Uvicorn, Pydantic 2.0 (Port 8787)	Standalone MCTS engine running 600+ rollouts across Markov disruption states.
AI & RAG Pipeline	Groq SDK (LLaMA 3.3 70B & 3.1 8B), Google Gemini 2.5 Flash, Zod	5-module parallel LLM reasoning pipeline with automatic rate-limit failover and schema validation.
Data Ingestion Feeds	AISStream (WebSocket), NewsAPI, EIA Open Data, API Ninjas	Real-time streaming telemetry: maritime vessels, geopolitical OSINT, crude and LNG spot prices.

3. END-TO-END INFORMATION FLOW: "WHAT FLOWS WHERE"

To understand how SSCA processes real-world chaos into actionable boardroom decisions, follow the exact 6-stage lifecycle below:

1. **Multi-Modal Ingestion & Telemetry Aggregation**
   Live vessel coordinates from `AISStream WebSocket`, breaking geopolitical news from `NewsAPI`, and spot commodity prices (Brent, WTI, LNG, Coal) from `EIA / API Ninjas` stream into the Next.js ingestion layer.

2. **Fact Extraction & Entity Resolution (Groq LLaMA 3.1 8B)**
   Raw unstructured OSINT text is passed to `LLaMA 3.1 8B` on Groq (sub-200ms latency) to extract named entities: disrupted chokepoints, involved nation-states, targeted vessel types, and military escalation flags.

3. **Knowledge Graph Grounding & BFS Topological Traversal**
   Entities are mapped against our **100+ node Strategic Trade Knowledge Graph**. The system executes a 3-hop Breadth-First Search (BFS) tracing: `Chokepoint (Hormuz) → Commodities (Crude Oil) → Import Dependencies → Downstream Refineries (Jurong / Jamnagar) → Power Grids → Alternatives`.

4. **Python Monte Carlo Tree Search (MCTS) Simulation**
   Next.js proxy (`/api/monte-carlo`) queries the Python FastAPI microservice (`:8787`). The MCTS engine simulates **600 iterations** across Markov states (Trigger → Triage → Rerouting → Resolution), returning dynamic recalibrated severity %, escalation risk %, and optimal mitigation policies (e.g. `reserve-logistics`).

5. **5-Agent Parallel AI Synthesis (Groq LLaMA 3.1 70B)**
   The pre-computed graph context, MCTS metrics, and corroborated news signals are dispatched concurrently to **5 specialized Groq AI agents**: (1) Executive Briefing, (2) Supply Chain Impact, (3) Policy Recommendations, (4) Scenario Forecasting, and (5) Corroborated Evidence Audit Trail.

6. **Zod Schema Validation & Digital Twin Presentation**
   The multi-agent output is assembled, filled for missing gaps from the deterministic graph, validated through a strict `Zod schema`, and rendered instantly onto the live MapLibre digital twin and interactive telemetry HUDs.

4. DEEP DIVE INTO THE MONTE CARLO (MCTS) ENGINE

Unlike conventional systems that guess risk using arbitrary static multipliers, SSCA's **Python Simulation Microservice** implements a true **Monte Carlo Tree Search algorithm** adapted specifically for multi-stage supply chain breakdowns.

Markov Disruption State Space:
`[ TRIGGER ] ──► [ TRIAGE ] ──► [ REROUTING ] ──► [ CAPACITY-REBUILD ] ──► [ RESOLUTION ]`

During each simulation run (600 rollouts in ~2.5ms), the MCTS engine evaluates:

- **Exploration vs. Exploitation (UCB1):** Balances testing unvisited strategies against exploiting high-reward responses: `reserve-logistics`, `strategic-rerouting`, `spot-market-arbitrage`, `diplomatic-escort-convoy`.
- **Dynamic Severity %:** Recalibrated true impact based on node visit frequencies and reward outcomes across trees.
- **Escalation Probability (%):** Statistical odds that regional military or diplomatic tension escalates into prolonged blockade.
- **Economic Stress Index (%):** Cumulative freight spot surge + war-risk insurance premiums + rerouting friction.
- **Zero-Downtime Fallback:** If the Python microservice is offline, the Next.js API proxy automatically returns a graceful fallback signal, allowing the dashboard to operate seamlessly on deterministic static baselines without crashing.

5. PLATFORM FEATURE MODULES BREAKDOWN

🌏 Geopolitical Risk & Intelligence

Multi-agent OSINT news monitoring and automated strategic briefing generation.

- 5 parallel Groq LLaMA 3.1 70B analysis modules
- Corroborated evidence confidence scoring (Strong/Moderate/Weak)
- Automatic Gemini 2.5 Flash failover router
- Interactive "Ask Intelligence" RAG chatbot

🗺️ Command Center Digital Twin

Interactive MapLibre GL geospatial command interface with live vessel tracking.

- Real-time AIS maritime tanker fleet streaming
- Global chokepoint threat radius heatmaps
- Trade corridor density & speed over ground metrics
- One-click country switcher (Singapore & India profiles)

🎮 Scenario Simulator & MCTS

What-if disruption stress testing with graph propagation & Monte Carlo.

- Preset scenarios: Hormuz Closure, Malacca Piracy, etc.
- Live Python MCTS panel with 600 rollout telemetry
- Interactive decision levers: SPR Drawdown, Route Diversion
- Side-by-side Delta comparisons (Baseline vs Levers)

📦 Adaptive Energy Procurement

Spot market intelligence, supplier risk matrix, and contract exposure.

- Live EIA & API Ninjas spot pricing (Brent, WTI, LNG, Coal)
- Supplier concentration Herfindahl-Hirschman scoring
- Alternative supplier match engine with freight penalties
- Contract vulnerability & force-majeure risk index

🔬 Refinery & Processing Cockpit

Downstream refinery capacity, crude diet allocation, and throughput monitoring.

- Refinery complex mapping (Jurong, Jamnagar, Paradip)
- Light sweet vs. heavy sour crude diet modulation
- Distillation unit utilization & margin stress modeling
- Product yield breakdowns (Jet Fuel, Gasoline, Diesel, Naphtha)

🛢️ Strategic Reserves & Analytics

Emergency petroleum reserve depletion models and cross-scenario analytics.

- Days of forward cover drawdown simulator
- SPR underground cavern fill gauges vs IEA mandates
- Cross-Scenario comparative supply deficit charts (Mtpa)
- Energy Dependency Network with Bezier flow streams

6. QUICK START & EXECUTION GUIDE FOR TEAMMATES

1. Prerequisites

Make sure your machine has **Node.js 18+** and **Python 3.10+** installed.

2. Installation & One-Command Boot

```
# 1. Clone repo & install Node.js dependencies
npm install

# 2. Install Python simulation microservice dependencies (fastapi, uvicorn, pydantic)
npm run mc:install

# 3. Create your .env.local file from the template
cp .env.example .env.local

# 4. Start the entire dual-stack platform simultaneously in one command!
npm run dev:full
```

This boots both services concurrently in one terminal:
• **Frontend Web Dashboard:** `http://localhost:3000` (Next.js 16)
• **Python MCTS Server:** `http://localhost:8787` (FastAPI Microservice)

7. HACKATHON PITCHING & DEMO STRATEGY (3-MINUTE FLOW)

Minute 1: The Hook & The Problem (Visual Shock)
Open the **Command Center**. Show the live AIS vessels streaming in real-time across the Strait of Malacca and Strait of Hormuz. Explain that 80% of world energy flows through these bottlenecks, yet supply chain teams still rely on reactive spreadsheets days after ships halt.

Minute 2: The Core Tech & AI Reasoning (Deterministic RAG)
Switch to **Geopolitical Intelligence**. Click "Generate Intelligence Report". Show how 5 Groq AI agents reason in parallel over our 100-node Knowledge Graph in seconds. Emphasize that the AI does *not* hallucinate—it traverses deterministic trade paths from choke point to refinery.

Minute 3: The Monte Carlo Disruption Engine (The Killer Feature)
Navigate to **Scenario Simulator**. Select "Strait of Hormuz Blockade". Point out the **Monte Carlo MCTS panel**: explain that our Python microservice ran 600 probabilistic rollouts to compute dynamic risk (54.4% severity, 21.9% escalation). Turn on the "Release Strategic Reserves" lever and show how the systemic supply gap drops instantly.

8. KEY DEFENSE POINTS FOR JUDGES / Q&A

JUDGE'S QUESTION	OUR TECHNICAL DEFENSE & ARCHITECTURE ANSWER
"How do you prevent LLM hallucinations on supply chain facts?"	We do not let LLMs guess trade routes. Our **Deterministic Knowledge Graph** runs a 3-hop BFS first, injecting exact trade dependencies into the prompt. The LLM only synthesizes the final briefing, which is strictly validated with **Zod**.
"Why use Monte Carlo (MCTS) instead of simple rule heuristics?"	Supply chain disruptions are Markov processes with dynamic branching (rerouting vs escorts vs diplomatic waivers). MCTS simulates 600 potential futures to discover non-obvious optimal actions that heuristics miss.
"Is this tied to only one country or global?"	Fully global. The architecture supports dynamic country profiling (Singapore, India, Regional Hubs) with localized energy import baskets, reserve mechanisms, and port topologies.
"What happens if external APIs or Python goes down?"	Zero downtime. The system features automatic dual-LLM fallback (Groq → Gemini) and seamless fallback to deterministic baselines if the Python server is offline.

Summary for Teammates: We have built an end-to-end, multi-agent, dual-stack digital twin that combines real-time data ingestion, deterministic graph algorithms, and probabilistic AI. Focus on the live demo and MCTS story during presentation!